

Student Feedback Survey



Machine Learning
Algorithms and
Applications

<https://go.blueja.io/KxnqfkWtL0i0FXpAAOwj5A>

- Completely anonymous
- All feedback are welcomed

The screenshot shows the login interface for the UTS online Student Feedback Survey (SFS). At the top, there's a banner with the survey title and a photo of students. Below the banner, a status bar shows 'Open online surveys: 42' and 'Current participation: 14.94%'. The main heading is 'Welcome to the UTS online Student Feedback Survey (SFS)'. It includes instructions for students and staff to enter their UTS ID and password. There are two login sections: one for 'STUDENT' and one for 'STAFF', each with input fields for 'UTS ID' and 'UTS password' and a 'GO' button. A 'SFS Privacy Notice' link is at the bottom. The footer contains 'UTSOnline', 'One Stop Student Admin', and the version 'v2.4.1'.

Machine Learning Algorithms and Applications

Lecture 8

(Block 3)

Plan for Today



Dimensionality Reduction



Anomaly Detection



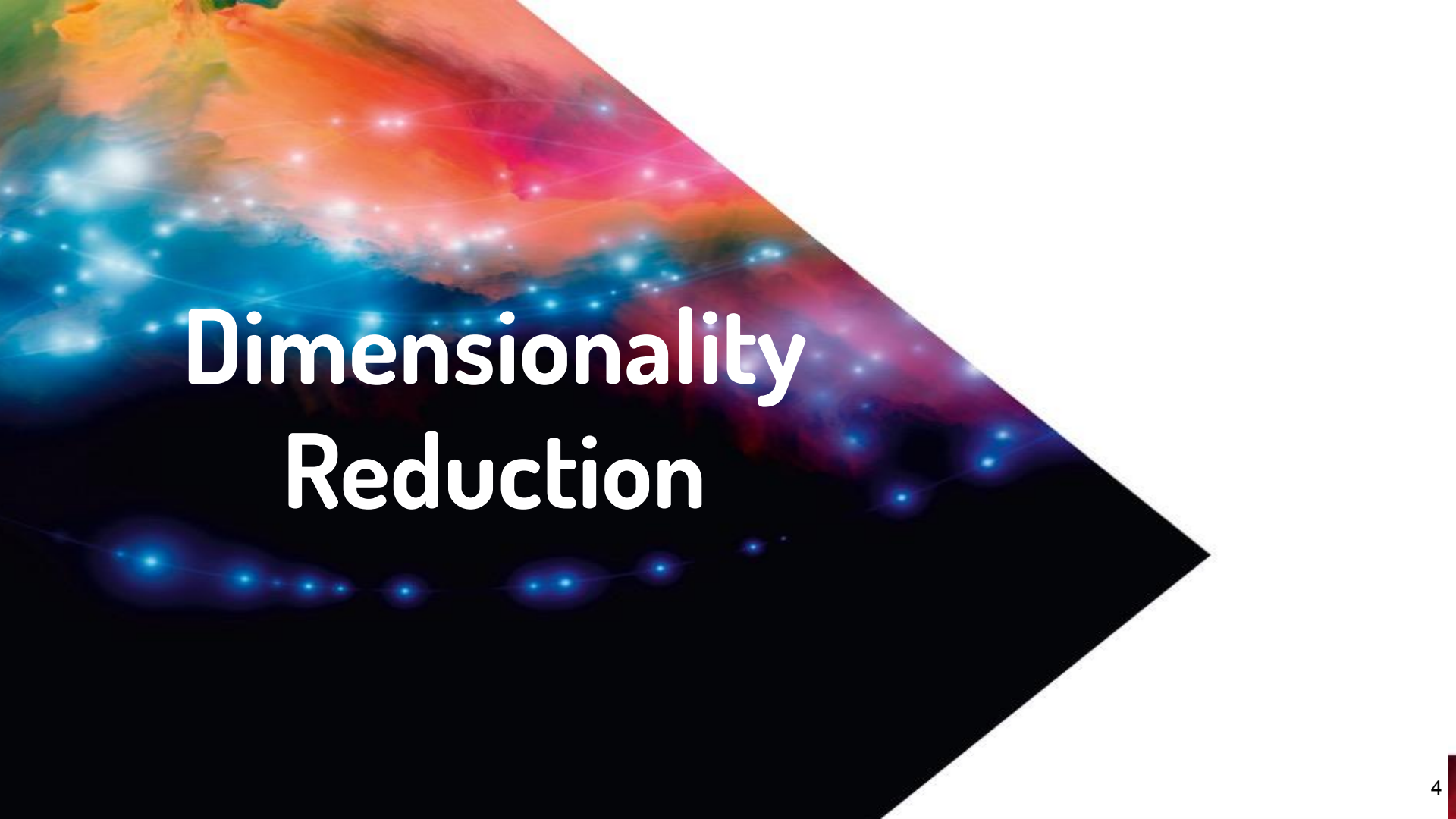
PCA



**IsolationForest
Local Outlier Factor**



Lab



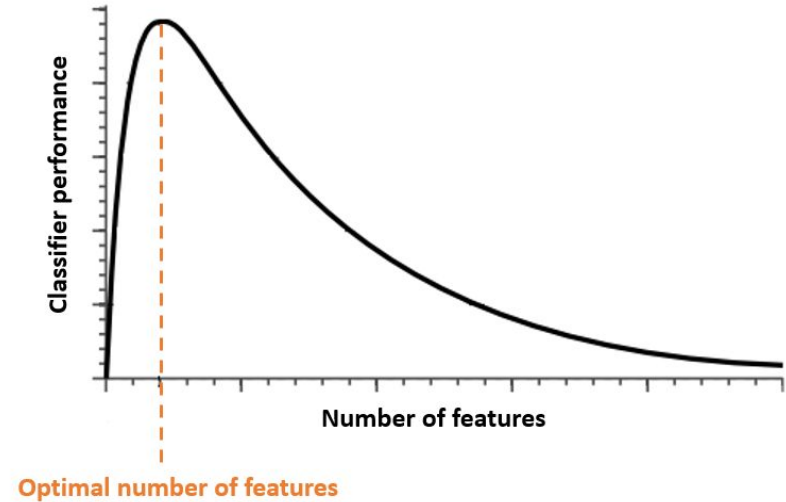
Dimensionality Reduction

Curse of Dimensionality

Dataset can sometimes contain a large number of features which can create additional problems while analysing or visualising data for instance.

They are usually referred as high dimensional data when they have more than hundreds of features.

While training machine learning models, large dataset will require more resources in order to process all the computations. It may also introduce too much noise for the algorithm to find relevant and meaningful patterns. Such problems are referred as 'Curse of Dimensionality'.



Data Sparsity

Sparse data situation occurs when columns doesn't contain much actual data. Sparse data can be empty or has a zero value.

Dense Matrix

1	2	31	2	9	7	34	22	11	5
11	92	4	3	2	2	3	3	2	1
3	9	13	8	21	17	4	2	1	4
8	32	1	2	34	18	7	78	10	7
9	22	3	9	8	71	12	22	17	3
13	21	21	9	2	47	1	81	21	9
21	12	53	12	91	24	81	8	91	2
61	8	33	82	19	87	16	3	1	55
54	4	78	24	18	11	4	2	99	5
13	22	32	42	9	15	9	22	1	21

Sparse Matrix

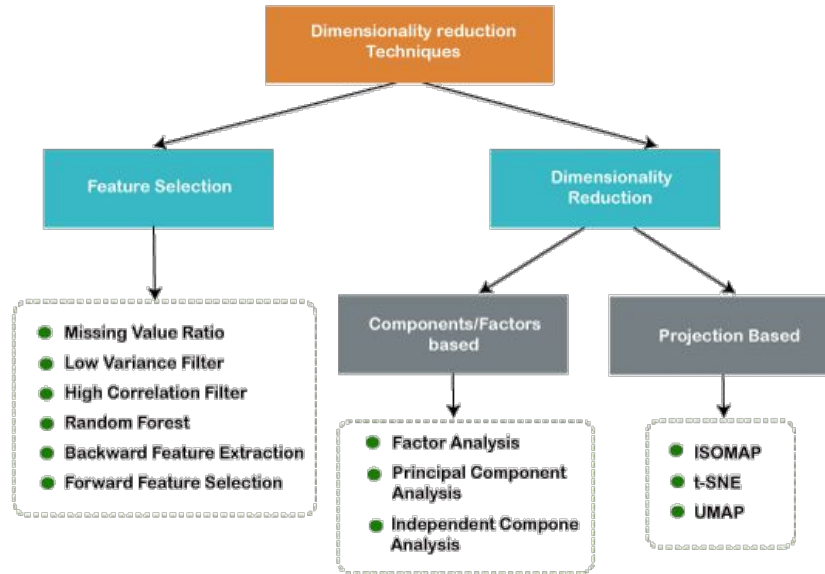
1	.	3	.	9	.	3	.	.	.
11	.	4	.	.	.	.	.	2	1
.	.	1	.	.	.	4	.	1	.
8	.	.	.	3	1	.	.	.	.
.	.	.	9	.	.	1	.	17	.
13	21	.	9	2	47	1	81	21	9
.	.	.	.	.	.	.	.	.	.
.	.	.	.	19	8	16	.	.	55
54	4	.	.	.	11	.	.	.	.
.	.	2	.	.	.	.	22	.	21

One-Hot Encoding creates sparse data by generating x new binary columns.



Dimensionality Reduction

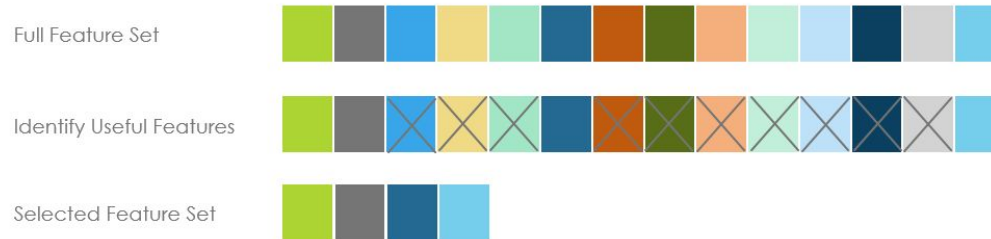
Dimensionality Reduction is the process of reducing the number of input variables in a dataset. The objective is to convert the high-dimensional variables into lower-dimensional variables without losing too much valuable information.



Feature Selection

One way to deal with high dimensional data is to select only the features that are useful. There are multiple techniques that can be used for this:

- **Low Variance filter:** Attributes that do not have much variance (almost constant value) may not contribute to the predictability of the model
- **High Correlation filter:** When there is a strong linear relationship between 2 features, one of them can be removed as they can be almost considered as duplicates,
- **Multicollinearity:** This situation happens when 3 or more features have a strong linear relationship even if no pairs of them have high correlation.
- **Feature Ranking:** Decision Tree models can rank the features based on their importance to the predictability of the model



Feature Extraction

It refers to a family of techniques that create new set of features built from the original feature set. The number of new features is lower than the number of the original ones:

- **Factor Analysis (FA):** features are grouped according to their correlations. In a particular group, all features will have a high correlation between them, but a low correlation with features from other group.
- **Principal Component Analysis (PCA):** technique which extracts a new set of correlated variables from an existing large set of variables.
- **Independent Component Analysis (ICA):** It assumes original features are linear mixtures of some unknown latent variables that are mutually independent.





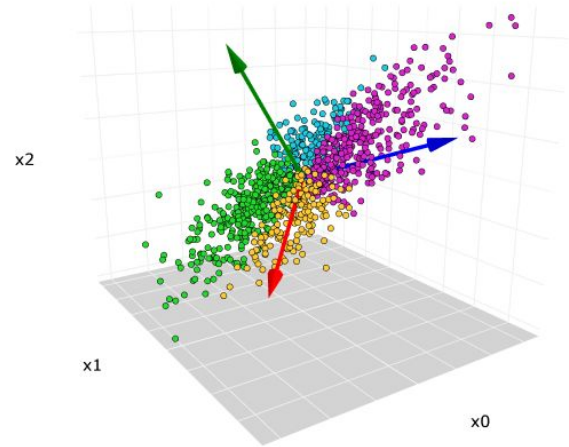
PCA

PCA

Principal component analysis (PCA) is a technique that transforms high-dimensions data into lower-dimensions while retaining as much information as possible.

*“PCA is defined as an orthogonal **linear transformation** that transforms the data to a **new coordinate system** such that the **greatest variance** by some scalar projection of the data comes to lie on the first coordinate (called the first principal component), the second greatest variance on the second coordinate, and so on.”*

Wikipedia



Variance

Variance refers to the spread of a data set. It's a measurement used to identify how far each observation in the data set is from the mean.

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

i	x
x0	17
x1	15
x2	23
x3	7
x4	9
x5	13

$$\bar{X} = 14$$

$$n = 6$$

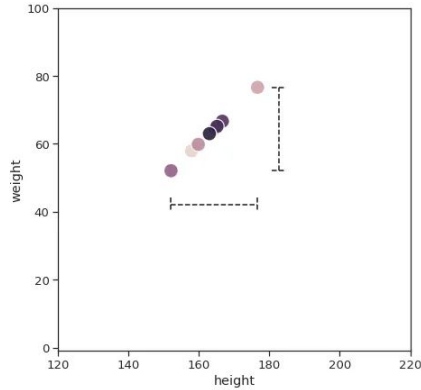


$$S^2 = \frac{166}{5} = 33.2$$

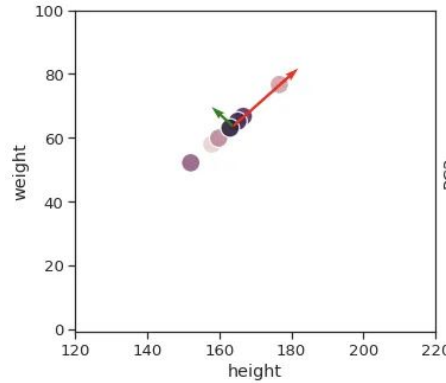


Feature Projection

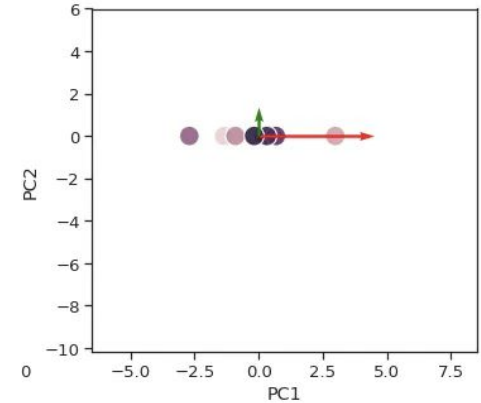
Projection methods seek to reduce the number of dimensions in the feature space whilst also preserving the most important structure or relationships between the variables observed in the data.



Original Data



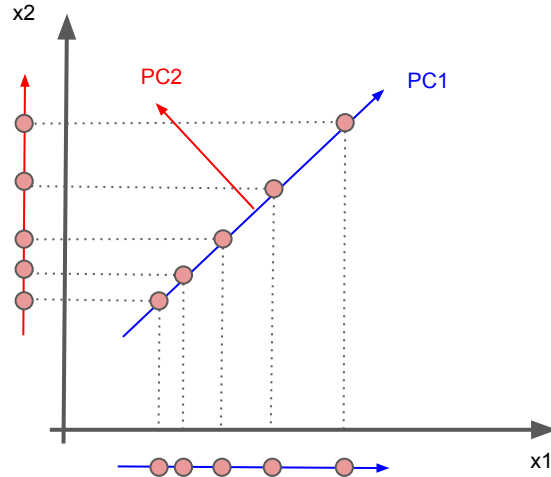
Identifying Principal Components



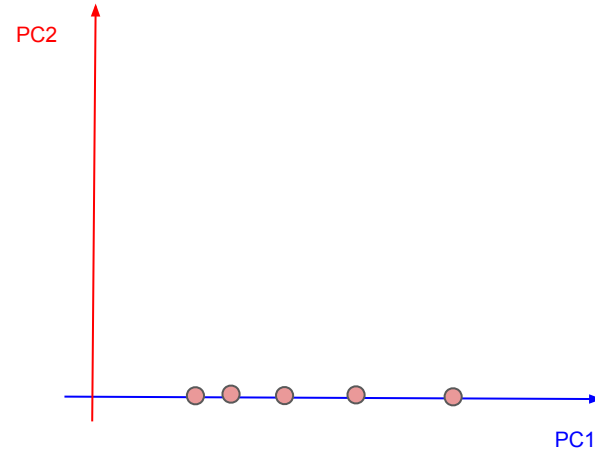
Projected data into Principal Components



Feature Projection and Variance



Original Data

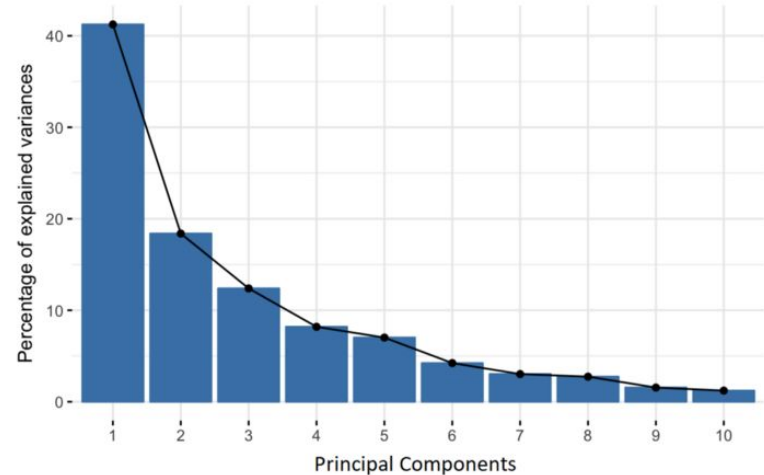
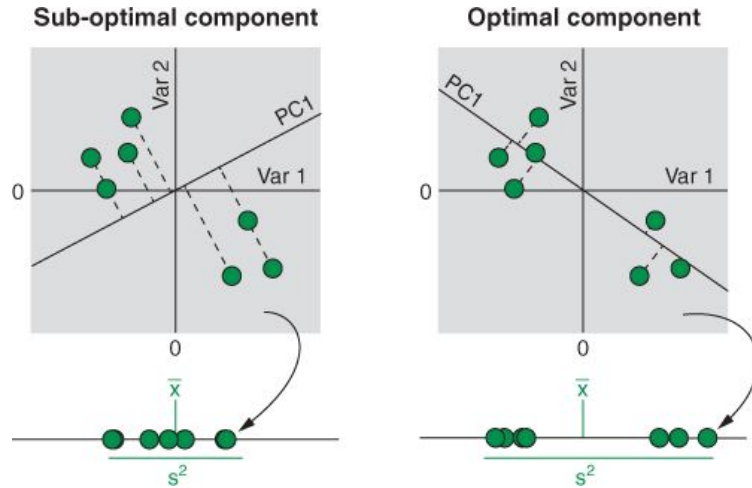


Projected data into Principal Components



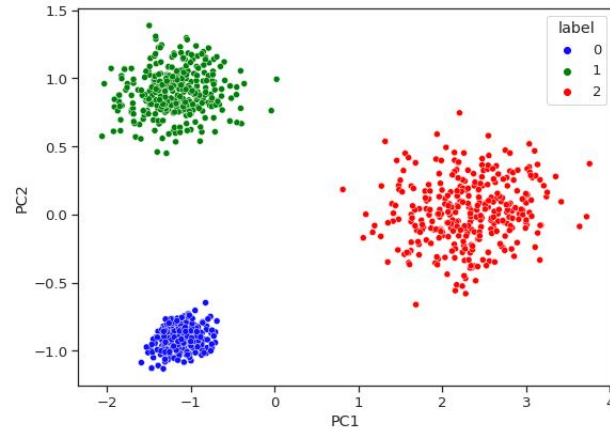
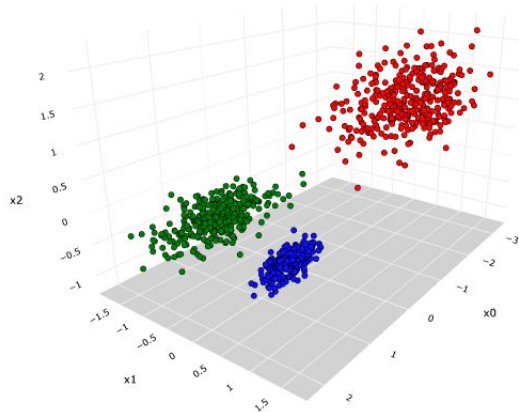
Principal Components

New variables that are constructed as linear combinations or mixtures of the initial variables maximising variance. These combinations are done in such a way that the new variables (i.e., principal components) are uncorrelated and most of the information within the initial variables is squeezed or compressed into the first components.



PCA for Visualisation

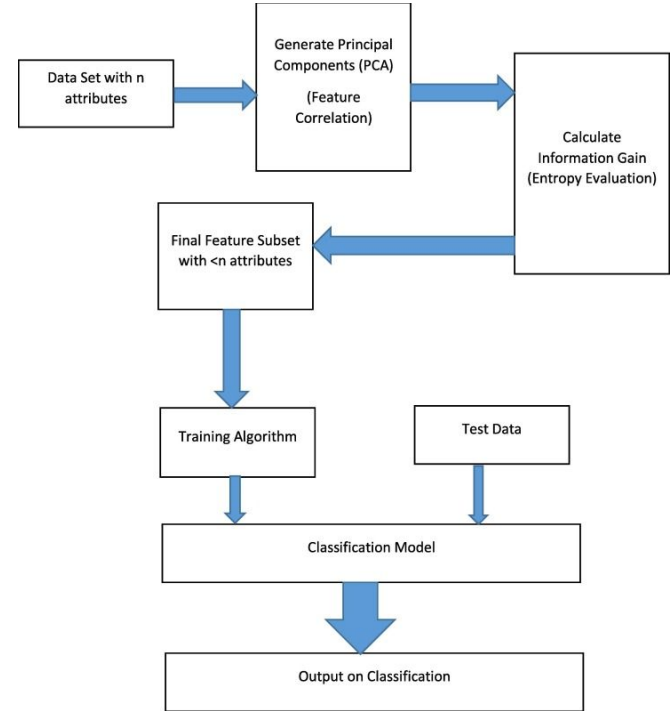
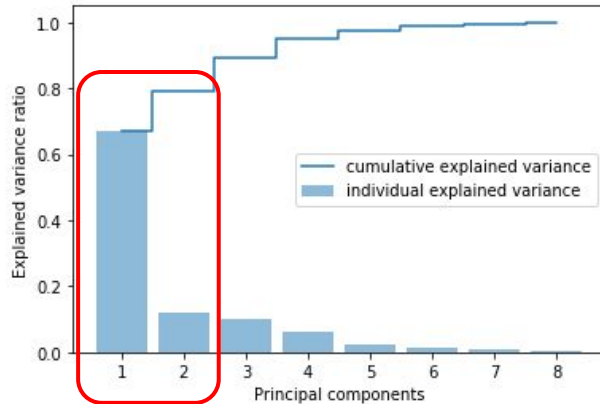
Instead of visualising data in multiple dimensions, we can use the principal components instead



PCA for ML

After finding the principal components, you can then select a subset of them and feed them to your machine learning model as inputs.

Suitable for both supervised and unsupervised problems.



Disadvantages

Independent variables are now less interpretable. PCA reduces your features into smaller number of components. Each component is now a linear combination of your original features, which makes it less readable and interpretable.

Information loss. Data loss may occur if you do not exercise care in choosing the right number of components.

Feature scaling. Because PCA is a *variance maximizing* exercise, PCA requires features to be scaled prior to processing.



Scikit-learn PCA

Generic API for sklearn models:



Import class

```
from sklearn.decomposition import PCA
```



Instantiate a model with hyperparameter

```
pca = PCA(n_components=2)
```



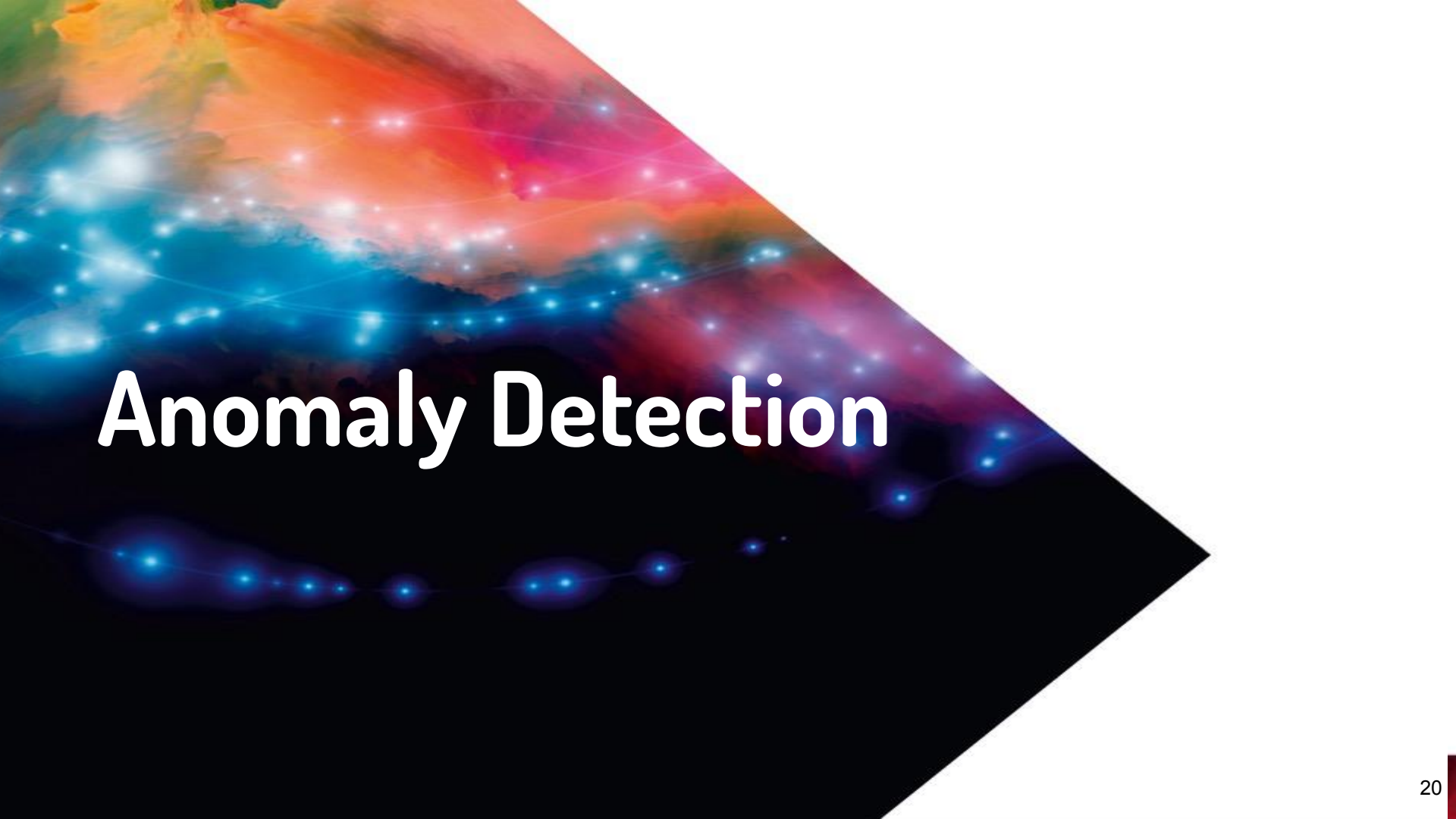
Fit a model

```
pca.fit(X)
```



Perform reduction

```
X = pca.transform(X)
```



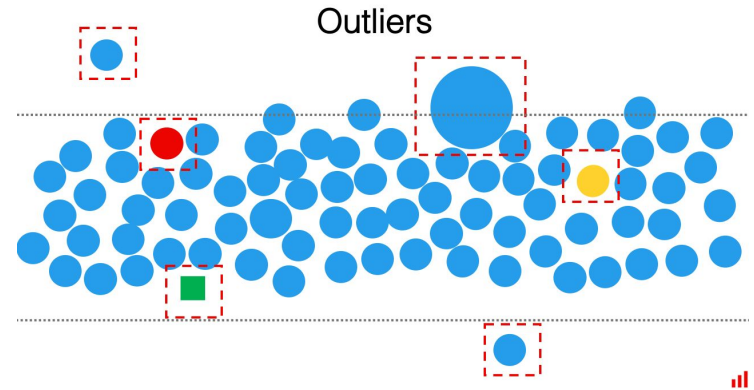
Anomaly Detection

Definition

Anomaly detection is the identification of rare events, items, or observations which are suspicious because they differ significantly from standard behaviors or patterns.

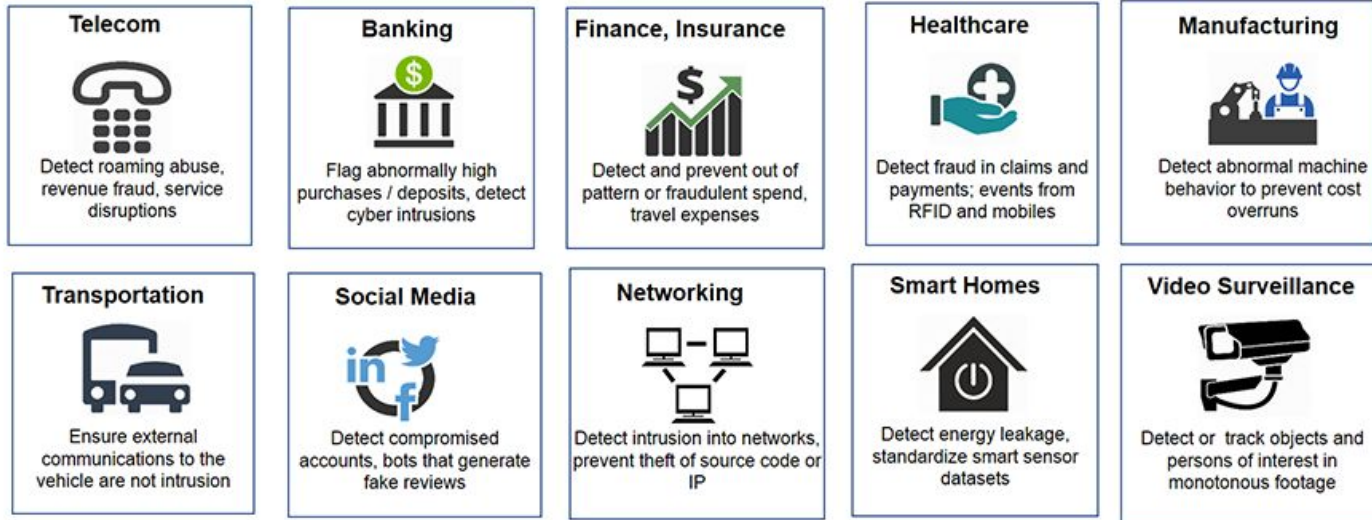
Anomalies in data are also called standard deviations, outliers, noise, novelties, and exceptions.

It is often applied on unlabeled data which is known as unsupervised anomaly detection.



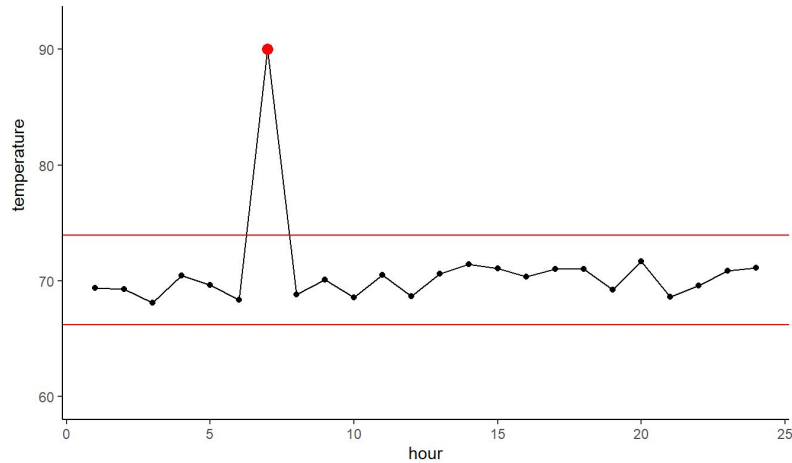
Use Cases

Anomaly detection is especially important in industries like finance, retail, and cybersecurity. It provides an automated way of detecting potentially harmful outliers.



What is an Anomaly

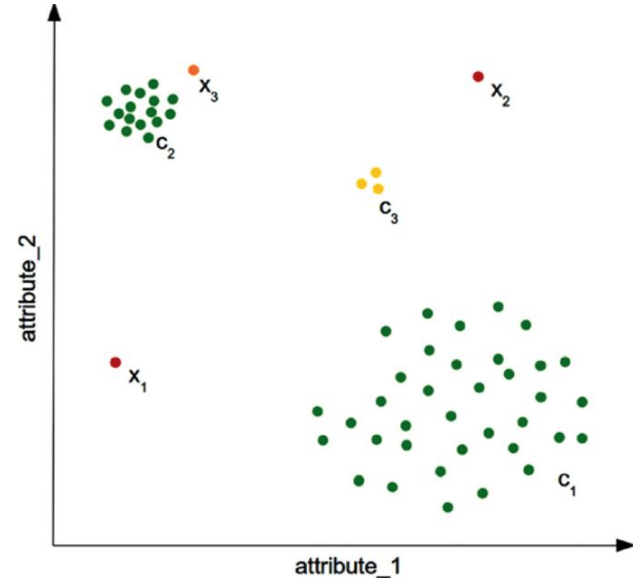
Any data point/observation that deviates significantly from the other observations and that occurs very rarely. Typically, anomalous data is linked to some sort of problem or rare event such as hacking, fraud, malfunctioning equipment, structural defects / infrastructure failures, or textual errors.



Anomaly versus Noise

In searching data for anomalies that are relatively rare, it is inevitable that the user will encounter relatively high levels of noise that could be similar to abnormal behavior.

This is because the line between abnormal and normal behavior is typically imprecise, and may change often as malicious attackers adapt their strategies.

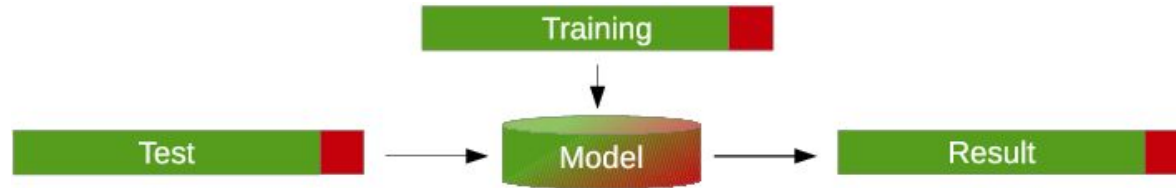


Supervised Anomaly Detection

(i.e. binary classification)

Build a predictive model with a labeled training set with normal and anomalous observations.

The advantage of supervised models is that they may offer a higher rate of detection than unsupervised techniques. This is because they can incorporate both data and prior knowledge, and find relevant patterns between variables.



(a) Supervised anomaly detection uses normal instances and anomalies for training and testing.

Unsupervised Anomaly Detection

Unsupervised methods do not demand manual labeling of training data. Instead, they operate based on the presumption that only a small, statistically different percentage of observations is malicious and abnormal. It is also called Outlier Detection.

These techniques thus assume collections of frequent, similar instances are normal and flag infrequent data groups as anomalies.



(c) Unsupervised anomaly detection uses no labeling information at all. The algorithm only takes intrinsic information of the data into account in order to detect anomalous instances being different from the majority.

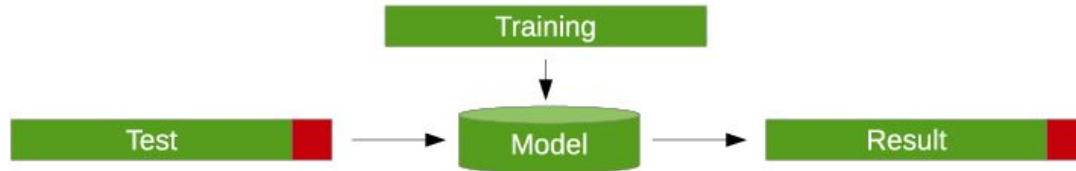


Semi-supervised Anomaly Detection

Semi-supervised anomaly detection refers to an approach of training a model on normal instances from a dataset. No anomalies are included in the training set.

The trained model will recognise either a new data observation is considered normal or outlier.

It is also referred as Novelty Detection.



(b) Semi-supervised anomaly detection requires solely normal data for training. The learned model detects then deviations in the test data from that norm.

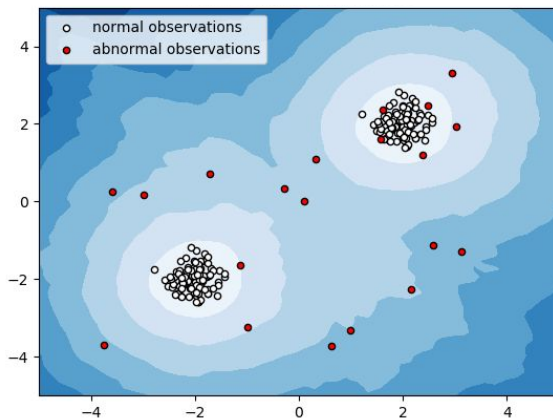


Local Outlier Factor

Local Outlier Factor (LOF)

Algorithm used for unsupervised outlier detection. It computes a score highlighting how different (density) is a data point compared against its neighbors.

When a point is considered as an outlier based on its local neighborhood, it is a **local outlier**. LOF will identify an outlier considering the density of the neighborhood.



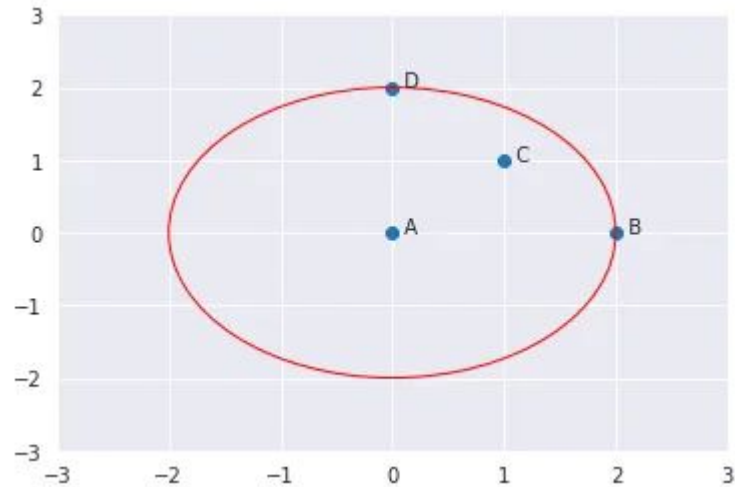
Distance and Neighbors

K-distance is the distance between a data point, and its K^{th} nearest neighbor.

K-neighbors includes a set of points that within the circle of radius K-distance.

Example:

If $K=2$, K-neighbors of A will be C, B, and D



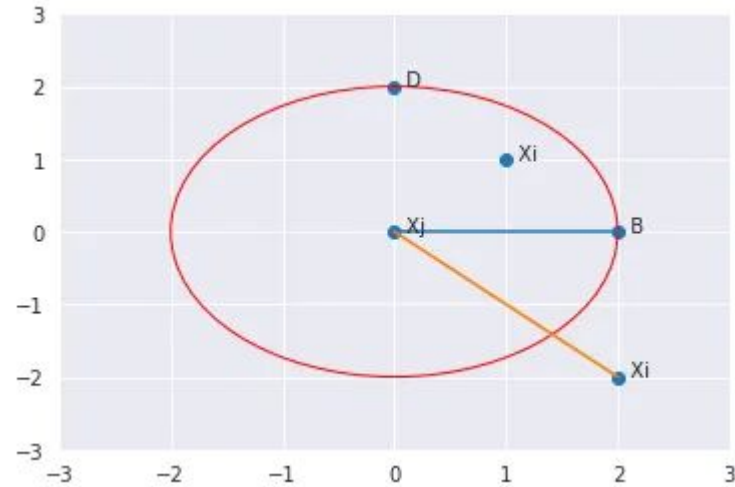
Reachability Density

It defines how easy it is to reach a point from another point.

If a point A is in the K-neighbourhood of B then it is easier to reach A and the reachability distance is lower. For a point A that is not in the K-neighbourhood of B, the reachability distance is higher.

It is defined as the maximum of the distance between two points and the k-distance of that point.

$$RD(X_i, X_j) = \max(K - \text{distance}(X_j), \text{distance}(X_i, X_j))$$



Local Reachability Density

It is inverse of the average reachability density of A from its neighbors.

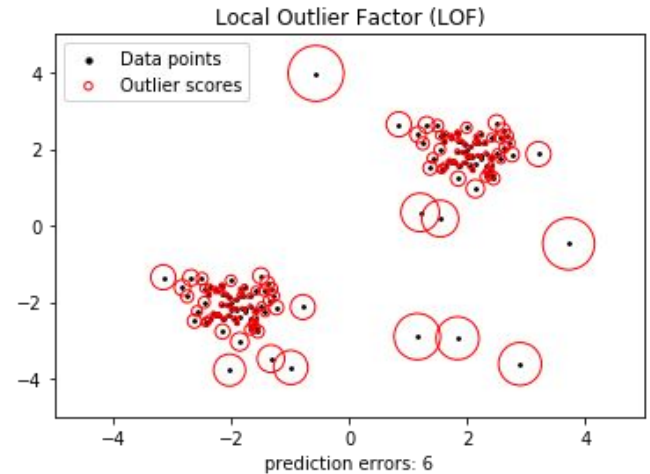
The higher the average reachability distance is (i.e., neighbors are far from the point), less density of points are present around a particular point.

The closer the points are, the distance is lesser, and the density is more.

$$LRD_k(A) = \frac{1}{\sum_{X_j \in N_k(A)} \frac{RD(A, X_j)}{\|N_k(A)\|}}$$

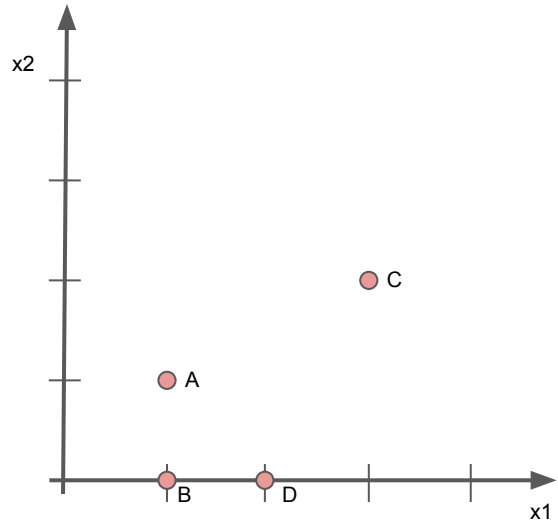
LOF greater than 1 => Anomaly

LOF close to 1 => Normal



L0F Step 1

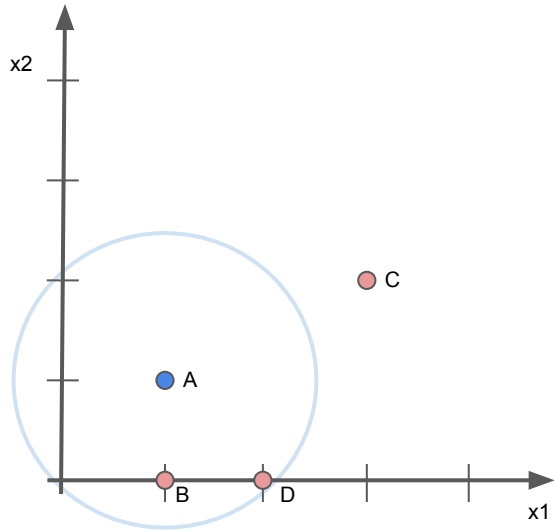
4 observations: A [1,1], B [1,0], C [3,2], D [2,0]



Calculate distance (Manhattan) between observations:

- $\text{distance}(A, B) = 1$
- $\text{distance}(A, D) = 2$
- $\text{distance}(B, D) = 1$
- $\text{distance}(A, C) = 3$
- $\text{distance}(B, C) = 4$
- $\text{distance}(C, D) = 3$

LOF Step 2



Find k-distance for $k=2$:

- $\text{distance_2}(A) = D = 2$
- $\text{distance_2}(B) = A \text{ or } D = 1$
- $\text{distance_2}(C) = A \text{ or } D = 3$
- $\text{distance_2}(D) = A = 2$

Find k-neighbors for $k=2$:

- $N(a) = \{b, d\}$
- $N(b) = \{a, d\}$
- $N(c) = \{a, d\}$
- $N(d) = \{a, b\}$

LOF Step 3

Calculate reachability distances:

- **reachabilty_distance(A,B)** = $\max\{\text{distance_2(B)}, \text{distance(A,B)}\} = \max\{1,1\} = 1$
- **reachabilty_distance(A,D)** = $\max\{\text{distance_2(D)}, \text{distance(A,D)}\} = \max\{2,2\} = 2$

- **reachabilty_distance(B,A)** = $\max\{\text{distance_2(A)}, \text{distance(B,A)}\} = \max\{2,1\} = 2$
- **reachabilty_distance(B,D)** = $\max\{\text{distance_2(D)}, \text{distance(B,D)}\} = \max\{2,1\} = 2$

- **reachabilty_distance(C,A)** = $\max\{\text{distance_2(A)}, \text{distance(C,A)}\} = \max\{2,3\} = 3$
- **reachabilty_distance(C,D)** = $\max\{\text{distance_2(D)}, \text{distance(C,D)}\} = \max\{2,3\} = 3$

- **reachabilty_distance(D,A)** = $\max\{\text{distance_2(A)}, \text{distance(D,A)}\} = \max\{2,2\} = 2$
- **reachabilty_distance(D,B)** = $\max\{\text{distance_2(B)}, \text{distance(D,B)}\} = \max\{1,1\} = 1$



LOF Step 4

Calculate the local reachability densities:

- **LRD(A)** = $1 / (\text{reachability_distance}(A,B) + \text{reachability_distance}(A,D)) / 2 = 2/3$
- **LRD(B)** = $1 / (\text{reachability_distance}(B,A) + \text{reachability_distance}(B,D)) / 2 = 1/2$
- **LRD(C)** = $1 / (\text{reachability_distance}(C,A) + \text{reachability_distance}(C,D)) / 2 = 1/3$
- **LRD(D)** = $1 / (\text{reachability_distance}(D,A) + \text{reachability_distance}(D,B)) / 2 = 2/3$



LOF Step 5

Calculate LOF Score:

- $\mathbf{LOF(A)} = ((\text{LRD(B)} + \text{LRD(D)}) / \text{LRD(A)}) / 2 = [(1/2 + 2/3) / (2/3)] / 2 = 7/8 = 0,875$
- $\mathbf{LOF(B)} = ((\text{LRD(A)} + \text{LRD(D)}) / \text{LRD(B)}) / 2 = [(2/3 + 2/3) / (1/2)] / 2 = 4/3 = 1,333$
- $\mathbf{LOF(C)} = ((\text{LRD(A)} + \text{LRD(D)}) / \text{LRD(C)}) / 2 = [(2/3 + 2/3) / (1/3)] / 2 = 4/2 = 2$
- $\mathbf{LOF(D)} = ((\text{LRD(A)} + \text{LRD(B)}) / \text{LRD(D)}) / 2 = [(2/3 + 1/2) / (2/3)] / 2 = 7/8 = 0,875$

$\text{LOF(C)} > 1 \Rightarrow \text{C is an anomaly}$



Scikit-learn LOF

Generic API for sklearn models:



Import class

```
from sklearn.neighbors import LocalOutlierFactor
```



Instantiate a model with hyperparameter

```
clf = LocalOutlierFactor(n_neighbors=20)
```



Fit a model and predict

```
y_pred = clf.fit_predict(X)
```

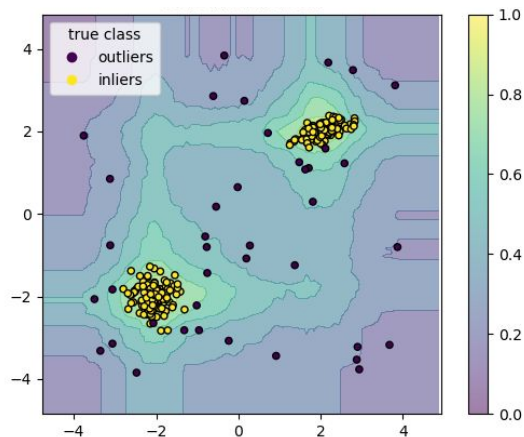
The background of the slide is a cosmic-themed image. It is split diagonally from the top-left corner to the bottom-right corner. The upper-left portion features a vibrant nebula with swirling clouds of orange, red, and pink, interspersed with numerous bright blue and white stars. The lower-right portion is a solid black field, punctuated by several horizontal trails of blue, out-of-focus stars. The text 'IsolationForest' is centered in the black area.

IsolationForest

IsolationForest

Unsupervised algorithm used for detecting outliers. Anomalies are considered as data points that are few and different from the rest.

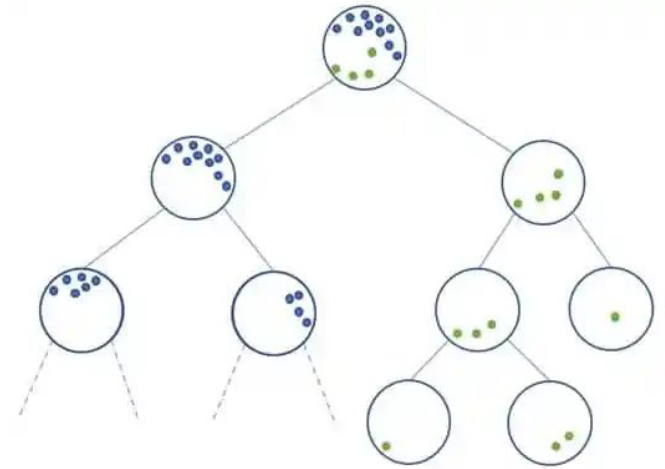
Isolation Forest is a tree-based model that will try to isolate anomalies on their own. The samples that travel deeper into the tree are less likely to be anomalies as they required more cuts to isolate them. Similarly, the samples which end up in shorter branches indicate anomalies as it was easier for the tree to separate them from other observations.



Pseudo-Algorithm

When given a dataset:

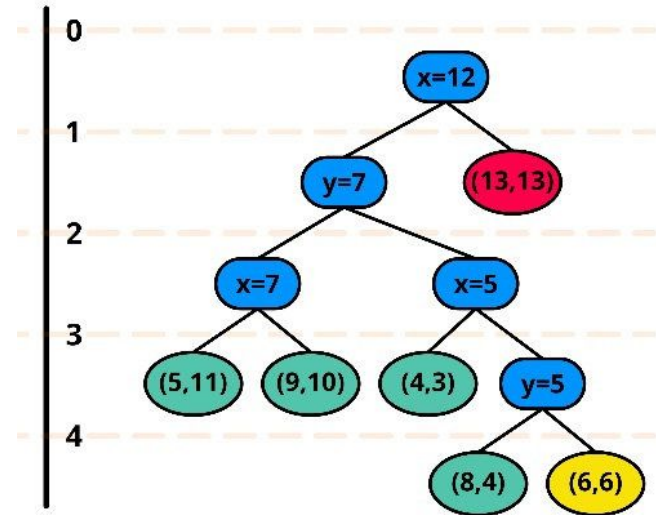
- Select a random subset of the data
- Until every point in the dataset is isolated:
 - a. Select one feature and a threshold randomly
 - b. Partition the data into left and right nodes using this feature and threshold



Isolation Anomaly

Isolation Forest recursively generates partitions on the dataset by randomly selecting a feature and then randomly selecting a split value for the feature.

Presumably the anomalies need fewer random partitions to be isolated compared to "normal" points in the dataset, so the anomalies will be the points which have a smaller path length in the tree, path length being the number of edges traversed from the root node.

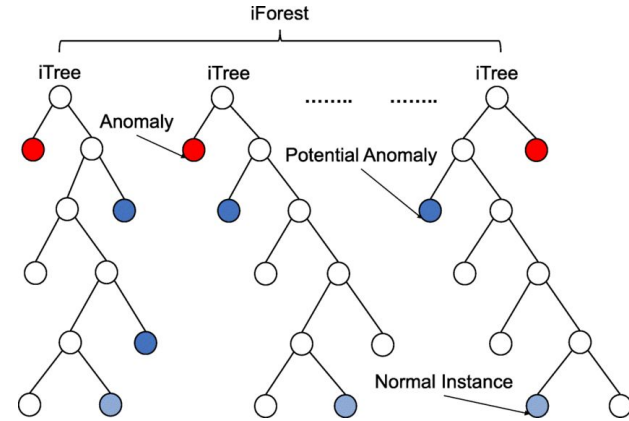


Building Forest

An 'anomaly score' can be assigned to each of the data points based on the depth of the tree required to arrive at that point.

The algorithm itself comprises of building a collection of isolation trees from random subsets of data, and aggregating the anomaly score from each tree to come up with a final anomaly score for a point.

Anomalous points will lead to short paths to leaves, *making them easier to isolate*, while interior points on an average will have a significantly longer path to the leaf.



Scikit-learn IsolationForest

Generic API for sklearn models:



Import class

```
from sklearn.ensemble import IsolationForest
```



Instantiate a model with hyperparameter

```
clf = IsolationForest(max_samples=100, random_state=0)
```



Fit a model and predict

```
y_pred = clf.fit_predict(X)
```



Time for Lab

